

- 4 (a) A point source of sound S emits a note of frequency 520 Hz at constant power. Two points along the wave X and Y, separated by a distance of 0.490 m, have a phase difference of $\frac{3}{2}\pi$ rad as shown in Fig. 4.1. Point X is 5.24 m away from S.

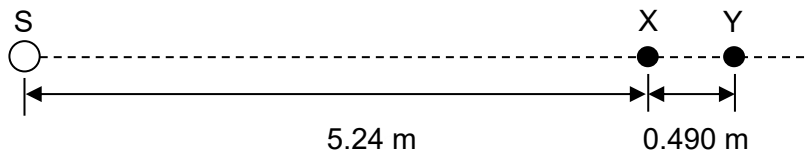


Fig. 4.1

- (i) Determine the wavelength and speed of the sound wave.

wavelength = m

speed = m s^{-1}

[3]

- (ii) Given that the intensity at X is $2.6 \times 10^{-3} \text{ W m}^{-2}$, calculate the intensity of the sound at Y.

intensity = W m^{-2} [2]

- (b) A parallel beam of light is incident on two polarisers P and Q. When the polarising directions are parallel, the amplitude of the emergent beam is A as shown in Fig. 4.2.

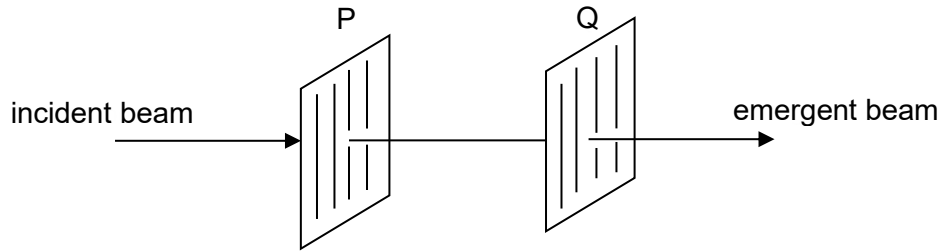


Fig. 4.2

- (i) Determine the angle through which Q must be rotated so that the amplitude of the emergent beam is reduced to $\frac{A}{2}$.

angle = ° [2]

- (ii) Determine the corresponding fractional change in the intensity of the emergent beam.

fractional change = [2]